Magnetism and electromagnetism

25 questions: 8 multiple choice, 17 structured.

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1 · Part 1

Paper 1 & 2 Units [1]

The rating plate on an electric drill reads "230 V 800 W".

Which quantity is measured in watts?

A current

B power

C voltage

D energy

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2 · Part 1

Paper 1 & 2 Magnetism [1]

One end of an unlabelled metal bar is held near the north pole of a bar magnet. That end of the bar is attracted to the magnet.

What can be concluded about the bar?

A It must be a magnet with a south pole at that end.

B It must be a magnet with a north pole at that end.

C It must be made of a non-magnetic metal.

D It is either a magnetic material or a magnet with a south pole at that end.

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3 · Part 1 Paper 1 & 2 Magnetism [1]

Which row describes a magnetically soft material?

	magnetised easily	keeps its magnetism when the magnet is removed
A	yes	no
B	yes	yes
C	no	yes
D	no	no

4 · Part 1 Paper 1 & 2 Electromagnetism [1]

Consider a horizontal wire that runs from west to east and carries a current from west to east. The wire is in a magnetic field that is directed vertically downwards.

Which direction is the force on the wire?

A south

B downwards

C north

D upwards

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5 · Part 1 Paper 2 only Electromagnetism [1]

An electron moves through a uniform magnetic field.

Which direction of motion gives no magnetic force on the electron?

A perpendicular to the field lines

B at 30° to the field lines

C parallel to the field lines

D at 60° to the field lines

6 · Part 1 Paper 1 & 2 Electromagnetism [1]

There is a force on a current-carrying wire placed at right angles to a magnetic field.

The current is increased and the direction of the magnetic field is reversed.

Which row describes the force on the wire now?

	size of force	direction of force
A	larger	reversed
B	larger	unchanged
C	unchanged	reversed
D	smaller	reversed

7 · Part 1 Paper 1 & 2 Electromagnetic induction [1]

When a bar magnet is pushed into a coil of wire, a voltage is induced. The coil is connected to a sensitive voltmeter, which shows a small reading while the magnet moves.

Which change increases the size of the induced voltage?

A using a coil with fewer turns

B pushing the magnet in more slowly

C holding the magnet still inside the coil

D using a stronger magnet

8 · Part 1 Paper 2 only Electromagnetic induction [1]

Which statement about transformers is correct?

A A transformer works with a steady direct current in its primary coil.

B A step-up transformer has more turns on its secondary coil than on its primary coil.

C The core is made of copper so that it conducts current from the primary coil to the secondary coil.

D A step-down transformer increases the power supplied to the output.

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9 · Part 1

Paper 1 & 2 Magnetism [5]

At a recycling centre, a large permanent magnet is used to separate steel cans from aluminium cans.

(a) State which type of can is attracted to the magnet. [1]

(b) The magnet picks up a steel can. While this can is touching the magnet, it attracts and holds a second steel can.

Explain why the first can is able to attract the second can. [2]

(c) Two bar magnets are brought close together.

State what happens to the magnets when

(i) a north pole is brought near a north pole [1]

(ii) a north pole is brought near a south pole. [1]

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10 · Part 1

Paper 1 & 2 Magnetism [6]

Three rods, P, Q and R, are made of different materials. One student tests them. The rods are the same size.

For each rod, the student holds one end against the same strong magnet and counts how many identical paper clips the other end can hold. The student then removes the magnet, waits one minute, and counts how many paper clips are still held.

rod	paper clips held while touching the magnet	paper clips held one minute after the magnet is removed
P	12	0
Q	9	8
R	0	0

(a) Identify the rod that is made of a non-magnetic material. [1]

(b) Identify the rod that is made of a magnetically hard material. Give a reason for your answer using the data. [2]

(c) A pick-up tool for pins is made from one of these rods. The rod must pick up pins while a magnet is held against one end, and drop all the pins as soon as the magnet is taken away.

Explain which rod should be used. [2]

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(d) State one variable, other than the size of the rods, that the student kept the same. [1]

11 · Part 1 Paper 1 & 2 Magnetism [5]

Magnetic fields are often represented by magnetic field lines.

(a) State what is meant by the term magnetic field line. [1]

(b) State the direction of the magnetic field lines outside a bar magnet. [1]

(c) Explain why two magnetic field lines can never cross. [2]

(d) State what is shown by magnetic field lines that are drawn close together. [1]

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12 · Part 1

Paper 1 & 2 | Magnetism | [6]

One student is given a bar magnet, a sheet of paper, a pencil and a small plotting compass. The student wants to find the magnetic field pattern around the bar magnet.

(a) Describe how the student should use this apparatus to plot the magnetic field lines around the magnet. [4]

(b) The student's field pattern is not symmetrical on the two sides of the magnet.

Suggest one reason for this. [1]

(c) The ends of the bar magnet are not labelled. The north-seeking end of the compass needle is marked.

State how the student can use the plotting compass to identify the north pole of the magnet. [1]

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13 · Part 1

Paper 1 & 2 | Magnetism | [6]

Two identical bar magnets are placed in a straight line on a bench with their north poles facing each other and a gap of 4 cm between them. A sheet of card is placed on top of the magnets. Iron filings are sprinkled onto the card and the card is gently tapped.

(a) Give a reason why the card is tapped. [1]

(b) Describe the pattern of the iron filings in the gap between the two north poles. [2]

(c) There is a point exactly halfway between the two north poles where there is no magnetic field.

Explain why there is no magnetic field at this point. [2]

(d) State one disadvantage of using iron filings, rather than a plotting compass, to investigate a magnetic field. [1]

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14 · Part 1

Paper 1 & 2 Magnetism [5]

Some experiments need a uniform magnetic field. A teacher has two flat permanent magnets, each with a north pole on one large face and a south pole on the opposite face.

(a) Describe how the teacher should arrange the two magnets to produce a uniform magnetic field between them. [2]

(b) Describe what the magnetic field lines look like in a uniform magnetic field. [2]

(c) The teacher uses the uniform field to investigate the force on a current-carrying wire.

Suggest why a uniform field is useful in this investigation. [1]

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15 · Part 1

Paper 1 & 2 | Electromagnetism | [6]

Two horizontal metal rails run straight away from a student. A light copper rod rests across the rails, so the rod runs from the student's left to the student's right. The rails are connected to a d.c. supply so that the current in the rod flows from left to right.

The student places a magnet so that the magnetic field at the rod is vertically upwards. When the current is switched on, the rod rolls along the rails.

(a) Explain why there is a force on the rod when there is a current in it. [2]

(b) Use the left-hand rule to predict the direction in which the rod rolls. [1]

(c) State two separate changes, each of which would make the rod roll in the opposite direction. [2]

(d) State one change that would increase the size of the force on the rod without changing its direction. [1]

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16 · Part 1

Paper 1 & 2 Electromagnetism [8]

The simple d.c. motor in this question has a rectangular coil of wire on an axle. The coil sits between the north pole of one magnet and the south pole of another magnet, so the magnetic field across the coil is horizontal. The two ends of the coil are connected to a split-ring commutator. Two carbon brushes press against the commutator and connect the coil to a battery and a switch.

At the start, the coil is horizontal, so its two long sides are at right angles to the magnetic field.

(a) Explain why the coil starts to turn when the switch is closed. [4]

(b) Explain the purpose of the split-ring commutator. [2]

(c) State two changes that would make the coil turn faster. [2]

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17 · Part 1

Paper 1 & 2 | Electromagnetism | [6]

Inside a loudspeaker there is a coil of wire attached to a paper cone. The coil sits in the strong magnetic field of a permanent magnet. An amplifier passes an alternating current through the coil.

(a) Explain how the alternating current makes the cone vibrate. [3]

(b) The frequency of the alternating current is 250 Hz.

State the frequency at which the cone vibrates. [1]

(c) The size of the alternating current is increased but its frequency is kept the same.

Explain how this changes the movement of the cone. [2]

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18 · Part 1 Paper 2 only Electromagnetism [7]

An electromagnet is used to hold a heavy fire door open. A steel plate on the door sticks to the electromagnet. When the fire alarm sounds, the current in the electromagnet is switched off and the door closes.

(a) Describe the construction of a simple electromagnet. [3]

(b) State two changes that would make the electromagnet stronger. [2]

(c) Explain why the core of the electromagnet is made of iron and not steel. [2]

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19 · Part 1

Paper 2 only

Electromagnetism

[6]

In a vacuum tube, a narrow beam of electrons travels horizontally from left to right across a vertical screen in front of a student. A magnetic field is applied to the beam. The field is directed into the screen, away from the student.

(a) Explain why the moving electrons in the beam are equivalent to an electric current. [2]

(b) Deduce the direction of the force on the electrons. Show your reasoning. [3]

(c) The electron beam is replaced by a beam of protons, also travelling from left to right in the same magnetic field.

Predict the direction of the force on the protons. [1]

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20 · Part 1

Paper 1 & 2 | Electromagnetic induction | [7]

The poles of a magnet face each other across a gap, so the magnetic field in the gap is horizontal. The student holds a straight wire horizontally in the gap, at right angles to the field. The ends of the wire are connected to a sensitive centre-zero voltmeter.

When the wire is moved vertically downwards through the gap, the pointer of the voltmeter moves to the right of zero.

(a) Describe what the voltmeter shows when

(i) the wire is held still in the gap [1]

(ii) the wire is moved vertically upwards through the gap at the same speed. [1]

(b) The student uses a motor to move the wire downwards through the gap at different steady speeds. The results are shown in the table.

speed of wire / m/s	induced voltage / mV
0.10	0.6
0.20	1.2
0.30	1.8
0.40	2.4

(i) Describe the relationship between the speed of the wire and the induced voltage. [2]

(ii) Predict the induced voltage when the wire moves at 0.50 m/s. [1]

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(c) The wire is then moved horizontally in the gap, directly from the north pole towards the south pole.

Explain why the voltmeter reading stays at zero. [2]

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21 · Part 1

Paper 1 & 2 | Electromagnetic induction | [8]

Inside a bicycle dynamo there is a small permanent magnet that is turned by the bicycle wheel. The magnet rotates inside a fixed coil of wire. The coil is connected to the bicycle's lamp.

(a) Explain why a voltage is induced in the coil when the magnet rotates. [2]

(b) Explain why the induced voltage is alternating. [2]

(c) The cyclist rides faster, so the magnet rotates faster.

Describe two ways in which the output voltage changes. [2]

(d) In a simple laboratory generator, a coil of wire rotates between the poles of a magnet.

State two factors, other than the speed of rotation, that affect the size of the voltage induced in the coil. [2]

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22 · Part 1

Paper 2 only

Electromagnetic induction

[7]

Transformers are used to change the size of an alternating voltage.

(a) Describe the structure of a transformer. [3]

(b) Explain how an alternating voltage across the primary coil produces a voltage across the secondary coil. [3]

(c) The coils of a transformer are made of insulated wire.

Give a reason why the wire must be insulated. [1]

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23 · Part 1

Paper 2 only

Electromagnetic induction [7]

The power supply in a school laboratory contains a transformer. The input to the primary coil is the 230 V a.c. mains supply. The output from the secondary coil is 12 V a.c. The primary coil has 1840 turns.

(a) State whether this is a step-up or a step-down transformer. [1]

(b) Calculate the number of turns on the secondary coil. [3]

(c) The current in the secondary coil is 2.3 A.

Calculate the current in the primary coil. Assume the transformer is 100% efficient. [3]

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24 · Part 1

Paper 2 only

Electromagnetic induction

[6]

Electrical energy from power stations is transmitted across the country through overhead cables at very high voltages, such as 400 000 V. Homes are supplied at about 230 V.

(a) Explain why electrical energy is transmitted through the cables at a very high voltage. [4]

(b) Explain why step-down transformers are used between the transmission cables and homes. [2]

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25 · Part 1

Paper 2 only

Electromagnetic induction

[8]

The turbines of an offshore wind farm produce a total electrical power of 160 MW. A transformer at the wind farm changes the voltage from 25 kV to 400 kV before the energy is sent to the shore. Assume the transformer is 100% efficient.

(a) Calculate the current in the secondary coil of the transformer. [3]

(b) Calculate the current in the primary coil of the transformer. [2]

(c) The primary coil has 750 turns.

Calculate the number of turns on the secondary coil. [2]

(d) A real transformer is not 100% efficient.

Suggest one way in which energy is wasted in a real transformer. [1]

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